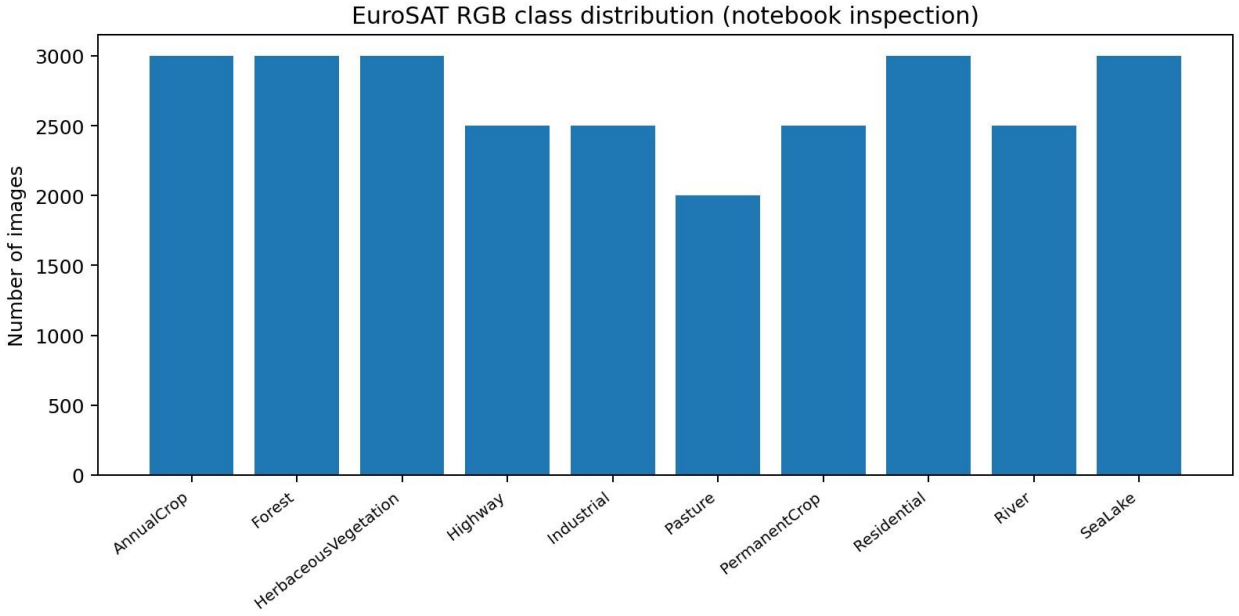
EuroSAT Land-Use / Land-Cover Classification Project

Dataset: EuroSAT RGB (Sentinel-2 satellite image patches)

Notebook workflow: dataset inspection, class distribution, visual sample review, image statistics, near-neighbour similarity, handcrafted-feature Random Forest, and ResNet18 transfer learning with random vs spatial split.



Important: Reported model scores are the outputs saved in the notebook. The ResNet18 experiment runs for only one epoch, so these are preliminary results—not a fully tuned benchmark.

1. Dataset overview

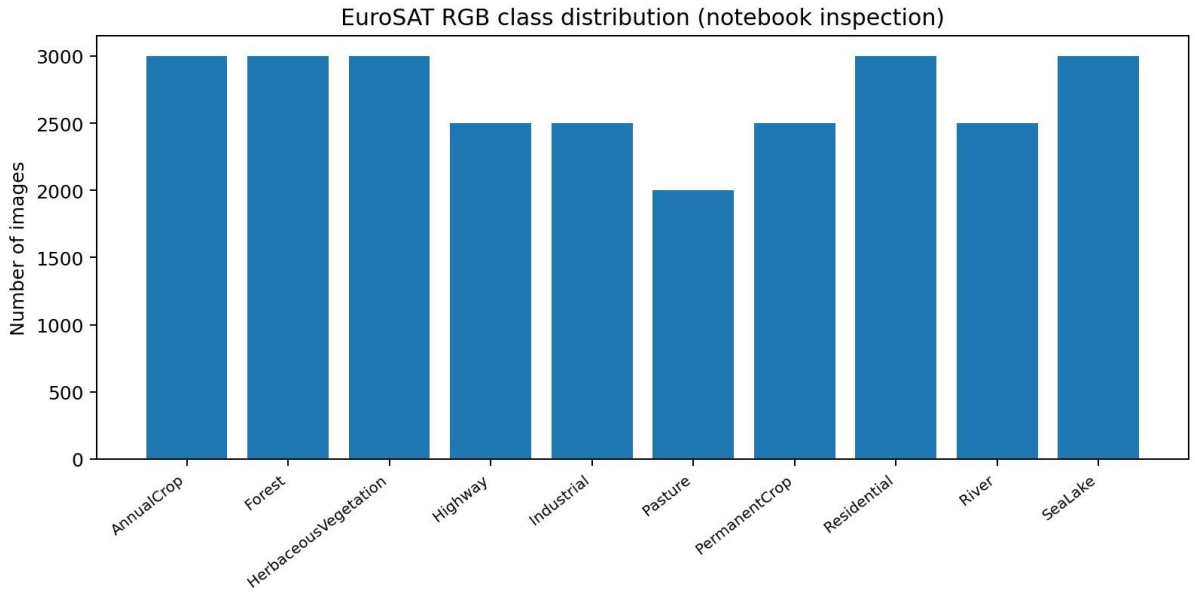
EuroSAT is a land-use and land-cover image dataset built from Sentinel-2 satellite imagery. The original dataset provides 27,000 labelled, geo-referenced image patches in 10 classes. The repository describes both an RGB version and a multispectral version with 13 spectral bands. This notebook uses the RGB directory version: 64 × 64 pixel, three-channel images.

Classes in the notebook: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, and SeaLake. Class counts range from 2,000 (Pasture) to 3,000 images per class.

2. Outputs and observations

2.1 Dataset integrity and balance

Notebook output: 27,000 rows and 15 computed metadata/feature columns; all images were 64 × 64 RGB. The exact-file MD5 duplicate check found 0 duplicates. This does not rule out near-duplicates or geographically related image patches.



2.2 Image statistics and visual inspection

The notebook samples eight images per class for visual inspection and computes RGB means/standard deviations, grayscale brightness, contrast, and a simple edge-energy statistic. In the saved output, SeaLake and Forest have low average contrast/edge energy, while Industrial and Residential have higher edge energy. These simple statistics can help explain class separability, but they are not a substitute for spectral or geographic analysis.

2.3 Near-neighbour similarity check

After resizing images to 16 × 16 and flattening RGB values, 1,596 images (5.9%) had a nearest-neighbour distance below the notebook's manually chosen threshold of 0.15. Among these cases, 97.8% of the nearest neighbours had the same class. This is a screening result only: the threshold is heuristic, and pixel similarity does not prove that samples overlap geographically.

3. Model experiments and saved results

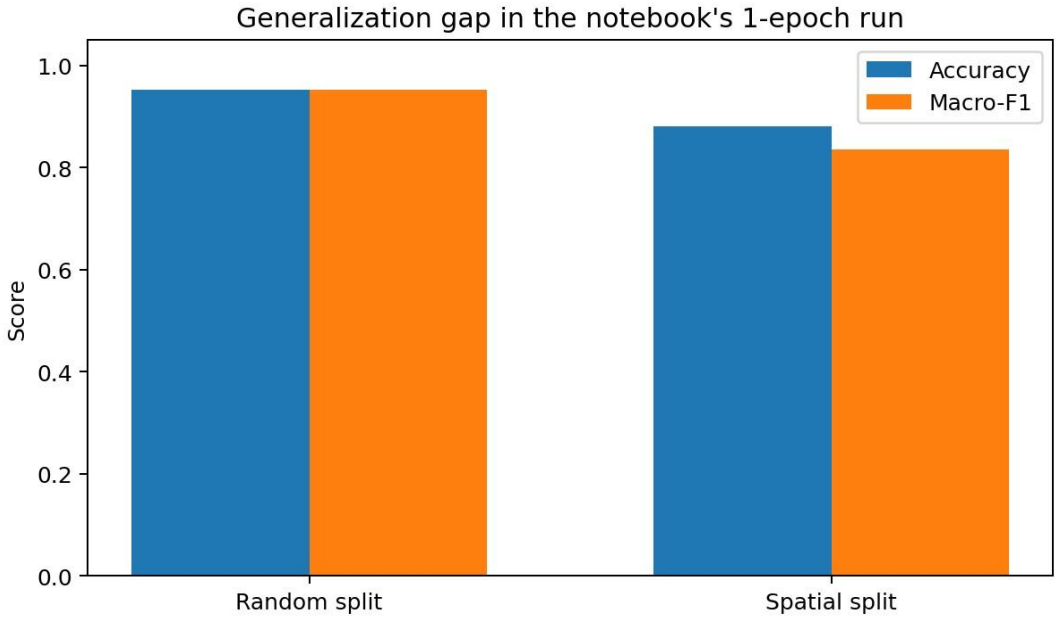
3.1 Handcrafted features + Random Forest

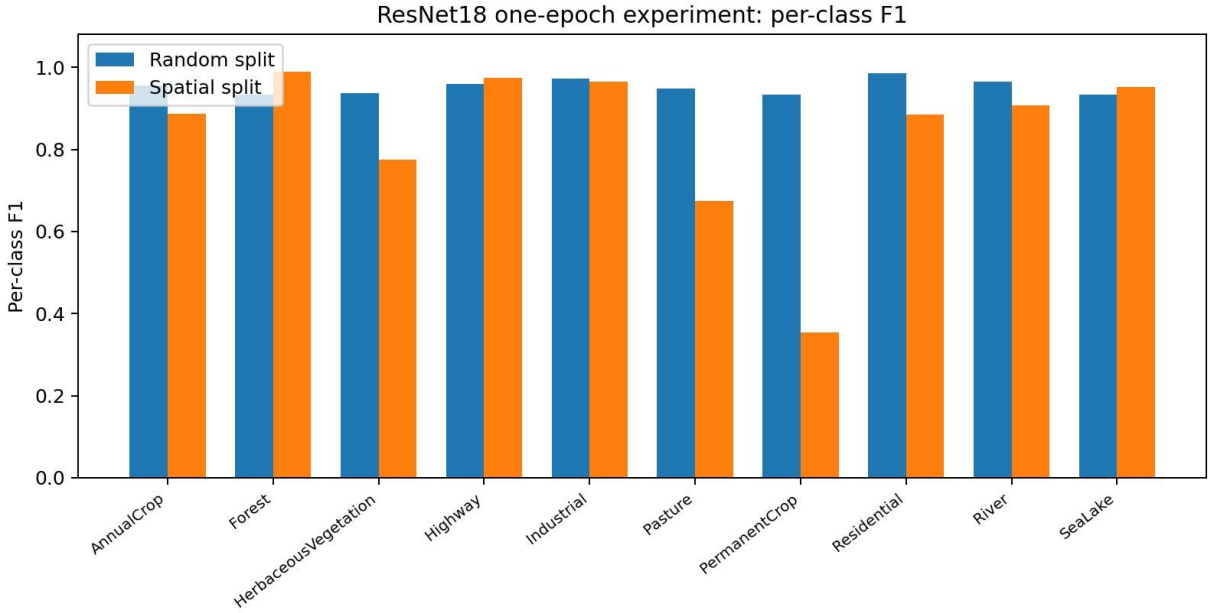
The notebook combines 16-bin histograms for each RGB channel with basic channel and image statistics, then trains a 300-tree Random Forest using an 80/20 stratified random split (random_state=0).

Test accuracy: 0.8467 (84.67%). Macro-F1: 0.837; weighted-F1: 0.844. Stronger F1 values include SeaLake (0.969), Forest (0.964), and Residential (0.930). Highway (0.574) and River (0.724) are more difficult for this feature set.

3.2 ResNet18 transfer learning: random vs spatial split

The notebook initializes ImageNet-pretrained ResNet18, replaces its final layer with a 10-class classifier, and trains for one epoch. The recorded outputs are: random split accuracy 0.9517 and macro-F1 0.9523; spatial split accuracy 0.8807 and macro-F1 0.8356.





The large difference suggests that evaluation with a spatial holdout is more challenging than a random split in this experiment. In particular, PermanentCrop F1 falls from 0.933 to 0.353, and Pasture from 0.949 to 0.674. Treat this as a strong preliminary signal, not definitive proof: the run is only one epoch, the spatial split implementation and geographic grouping should be checked, and training/evaluation settings must be controlled.

4. Research gaps and proposed contribution

The points below are **research opportunities inferred from the dataset properties and this notebook's preliminary results**; they should not be presented as proven gaps in all prior literature without a systematic literature review.

Gap 1 — Random-split performance may overstate geographic generalization

Motivation: The notebook's one-epoch ResNet18 run has 95.17% accuracy on the random split versus 88.07% on the spatial split; macro-F1 drops from 95.23% to 83.56%. Related satellite patches can be visually or geographically correlated, making random image-level evaluation potentially optimistic.

Research question: How much do common land-cover classifiers lose when tested on geographically separated areas rather than a random holdout? **Plan:** compare random, location/group-based, and spatial splits; document the split rules; repeat with multiple seeds; report accuracy, macro-F1, per-class F1, and confidence intervals.

Gap 2 — Underuse of all 13 spectral bands in RGB-only experiments

Motivation: This notebook uses RGB images, although EuroSAT also provides 13-band multispectral data. RGB may discard information useful for separating visually similar classes.

Research question: Do multispectral bands improve spatially held-out performance enough to justify additional model complexity? **Plan:** compare RGB, selected-band combinations, and all 13 bands using consistent splits and comparable model capacity. Fit normalization parameters on training data only.

Gap 3 — Class-specific failures need systematic explanation

Motivation: Random Forest F1 is particularly low for Highway (0.574), River (0.724), and PermanentCrop (0.745). The ResNet spatial-split result also shows a large PermanentCrop drop.

Research question: Which classes and image characteristics cause the largest generalization failures? **Plan:** report confusion matrices and per-class precision/recall/F1; inspect false positives/negatives; group errors by brightness, contrast, season/region if metadata supports it; test whether augmentations or spectral features help.

Gap 4 — Robustness to image conditions and distribution shifts

Research question: How stable are models under blur, reduced resolution, brightness/contrast changes, or band ablation? **Plan:** define realistic perturbation levels, evaluate all models on the same held-out set, and report performance degradation rather than only clean accuracy.

Gap 5 — Reproducible and statistically fair comparisons

Motivation: A one-epoch run is useful for a smoke test but not sufficient to claim a strong model comparison. The notebook also uses a particular reflectance scaling assumption in preprocessing that should be verified against the actual data format.

Plan: train to convergence or use a documented early-stopping rule; fix random seeds; tune hyperparameters only on validation data; compare against a simple baseline and published benchmarks; record library versions, split files, preprocessing, runtime, and repeated-run variability.

5. Suggested undergraduate research title

“Evaluating Geographic Generalization and Spectral Information in EuroSAT Land-Cover Classification”

This title gives a measurable contribution: compare RGB and multispectral inputs under both random and spatial evaluation, then analyze which classes lose the most performance. It builds directly on the notebook's strongest preliminary observation.

6. Limitations and next steps

1. Run the complete experiment for multiple seeds and sufficient epochs; the current ResNet scores are one-epoch outputs. 2. Verify the meaning, scale, and band order of inputs returned by the dataset loader. 3. Ensure spatial split boundaries prevent related locations from leaking across train and test. 4. Add a validation split and tune only on validation data. 5. Save the full classification reports, confusion matrices, training curves, and exact environment versions.

7. References

1. Helber, P., Bischke, B., Dengel, A., & Borth, D. (2019). “EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification.” IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 12(7), 2217–2226. DOI: 10.1109/JSTARS.2019.2918242.

2. EuroSAT repository and dataset description: https://github.com/phelber/EuroSAT

3. TensorFlow Datasets, EuroSAT catalog (RGB and 13-band variants): https://www.tensorflow.org/datasets/catalog/eurosat

4. Brigato et al. (2021). “Tune It or Don’t Use It: Benchmarking Data-Efficient Image Classification.” ICCV Workshops. Included here as related context on EuroSAT as a small-image, multispectral benchmark.